

Answers: Mean vs Median: Deal or No Deal

On the show *Deal or No Deal*, contestants choose from 26 suitcases. These are the amounts inside them (\$0.01 is one cent):

\$0.01	\$1	\$5	\$10	\$25	\$50	\$75	\$100	\$200
\$300	\$400	\$500	\$750	\$1,000	\$5,000	\$10,000	\$25,000	\$50,000
\$75,000	\$100,000	\$200,000	\$300,000	\$400,000	\$500,000	\$750,000	\$1,000,000	

a) What is the median amount?

The amounts are already in order. There are $n = 26$ amounts, an even number, so the median is the average of the 13th and 14th values:

$$\text{median} = \frac{750 + 1000}{2} = \$875$$

b) What is the mean amount?

$$\bar{x} = \frac{\sum x}{n} = \frac{3,418,416.01}{26} \approx \$131,477.54$$

c) Are the median and the mean close to each other? Why or why not?

No. The mean (\$131,477.54) is about 150 times the median (\$875). A few very large amounts (\$500,000, \$750,000, \$1,000,000) pull the mean up. The median is not affected by how large those values are.

d) What proportion of the prize amounts are bigger than the mean?

Only 6 amounts are bigger than \$131,477.54 (\$200,000 through \$1,000,000):

$$\frac{6}{26} \approx 0.23 \text{ (23\%)}$$

e) What proportion of the prize amounts are bigger than the median?

13 amounts are bigger than \$875 (\$1,000 through \$1,000,000):

$$\frac{13}{26} = 0.5 \text{ (50\%)}$$

The median always splits the data in half. The mean does not.